

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>
a) heat = mass × specific heat × temp change = 1 × 1 × (6 0 − 3 0) = 3 0
for water : 3 0 × 4 . 1 8 = 1 2 5 . 4 k cal (high temp is end)
b) heat for al = mass × specific heat × temp change = 1 × . 2 1 5 × (6 0 − 3 0) = 6 . 4 5
total heat = 1 3 1 . 8 + 6 . 4 5 = 3 0 Kcal
</think>
<answer>
G . 3 0 Kcal , 6 . 4 5 Kcal
</answer>